AI/ML intern DAY 20

OVERALL PROJECT UNDERSTANDING AND SYSTEM ARCHITECTURE ANALYSIS

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Introduction

After carefully studying the complete project architecture and workflow diagrams, I understood that the Raritone platform is much more than a Virtual Try-On application. It is an integrated AI-powered fashion ecosystem where multiple technical teams work together to provide an intelligent shopping experience.

Initially, I believed that Artificial Intelligence was used mainly for virtual garment fitting. However, after analysing the overall workflow, I realized that AI is deeply connected with backend services, data analytics pipelines, cloud infrastructure, recommendation systems, and business intelligence modules.

This research helped me understand how every department contributes to building a scalable and efficient fashion commerce platform.

Overall System Architecture

The platform operates through the collaboration of several independent modules that continuously exchange information.

The major components include:

• Vendor Management System

• Product Management Module

• Customer Application

• AI Processing Engine

• Backend Services

• Analytics Dashboard

• Payment and Logistics Services

• Cloud Storage Infrastructure

Every component performs a specific responsibility while remaining connected through APIs and centralized databases.

Backend System Understanding

The backend infrastructure acts as the operational core of the entire platform.

The major backend responsibilities include:

- User registration and authentication.
- Product catalogue management.
- Inventory tracking.
- Order processing.
- Vendor verification.
- Payment integration.
- Delivery management.
- API communication.
- Cloud resource management.

The backend ensures that data generated by customers, vendors, and AI modules remains synchronized across the platform.

Backend Technologies Studied

Technology	Purpose
Java	Server-side development
Spring Boot	API development
REST APIs	Service communication
PostgreSQL	Database management
MySQL	Relational storage
JWT	Secure authorization
Firebase Authentication	User login management
Docker	Containerization
AWS Cloud	Hosting and storage

Understanding the Data Analytics Team

The Data Analytics team transforms platform data into meaningful business insights.

The workflow indicates that every customer interaction generates valuable information that can later be analysed to improve business performance.

The analytics system monitors:

- Customer activity.
- Product popularity.
- Sales performance.
- Inventory movement.
- Vendor growth.
- Marketing effectiveness.
- Return statistics.
- Customer feedback.

These insights support decision-making across the organization.

Analytics Technologies

• Python

• Pandas

• NumPy

• SQL

• Power BI

• Tableau

• Matplotlib

• Seaborn

AI/ML Team Responsibilities

The Artificial Intelligence team enhances the platform by providing intelligent automation and personalization.

The AI modules researched include:

Virtual Try-On

- Human detection.

- Body segmentation.
- Pose estimation.
- Landmark extraction.
- Garment alignment.
- Image synthesis.

Fashion Recommendation

- User preference analysis.
- Personalized outfit suggestions.
- Similar product recommendation.

Business Intelligence

- Demand forecasting.
- Customer behaviour prediction.
- Return prediction.
- Trend analysis.

Vendor Assistance

- AI-generated descriptions.
- Product tag generation.
- SEO content creation.

AI Technologies Explored

• OpenCV

• MediaPipe

• OpenPose

• TensorFlow

• PyTorch

• VITON

• HR-VITON

• Diffusion Models

• Hugging Face Models

Data Requirements for AI Systems

To build intelligent fashion applications, the AI modules require multiple forms of data.

Important datasets include:

- Product images.
- Human pose datasets.
- Garment segmentation datasets.
- Fashion recommendation datasets.
- Customer behaviour data.
- Inventory history.
- Sales records.
- Product metadata.

Proper data organization significantly improves model performance.

Cross-Team Collaboration

One of the most important observations during this research was the level of interaction between teams.

The Backend Team manages business operations.

The Data Analytics Team extracts useful insights.

The AI/ML Team develops intelligent features.

All three departments depend on each other to maintain the overall platform.

Without backend services, AI cannot access live data.

Without analytics, business optimization becomes difficult.

Without AI, the platform loses personalization and intelligent automation.

Common Technologies Shared Across Teams

Technology	Teams Using It	Purpose
Python	AI/ML, Analytics	Machine Learning & Data Processing
SQL	Backend, Analytics	Database Management
AWS	Backend, AI/ML	Cloud Infrastructure
REST APIs	Backend, AI/ML	System Communication
Git & GitHub	All Teams	Version Control

Key Understanding

The complete workflow begins when vendors upload products to the platform.

The backend validates and stores this information.

AI systems process the product images and generate intelligent outputs.

Customers interact with these services through the application.

Every action generates additional business data that is collected and analysed to continuously improve the platform.

This creates a feedback cycle that combines Artificial Intelligence, Backend Engineering, and Data Analytics into a unified architecture.

Conclusion

This research helped me understand that Raritone is a complete AI-powered fashion commerce ecosystem rather than a standalone Virtual Try-On application.

The Backend team provides infrastructure and business logic, the Data Analytics team converts raw information into valuable insights, and the AI/ML team develops intelligent systems that improve personalization and user experience.

Studying the interaction between these departments provided a deeper understanding of large-scale software architecture and demonstrated how multiple engineering disciplines work together to build modern AI-driven platforms.